

MUHAMMAD AHMAD FAIZAN

AI & Backend Engineer

Python • FastAPI • AI Systems • AI Agents • RAG • Automation

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SUMMARY

AI & Backend Engineer specializing in designing and building production-ready AI systems, scalable backend services, and intelligent automation solutions. Experienced developing AI agents, multi-agent systems, Retrieval-Augmented Generation (RAG) pipelines, and RESTful backend architectures using Python, FastAPI, LangChain, LangGraph, and PostgreSQL. Delivered AI-powered applications and automation solutions for local and international clients across e-commerce, finance, and business operations. Passionate about building clean, scalable, and maintainable software that transforms AI capabilities into real business value.

CORE EXPERTISE

AI Systems • AI Agents • Multi-Agent Systems • Backend Architecture • Backend Development • Retrieval-Augmented Generation (RAG) • FastAPI • REST API Development • LLM Integration • Workflow Automation • Intelligent Automation • Software Architecture • Database Design

EDUCATION

Bachelor of Science in Software Engineering

2022 – 2026

National Textile University, Faisalabad

Built a strong foundation in software engineering, backend development, software architecture, databases, object-oriented programming, algorithms, data structures, software testing, and modern software development practices. Applied artificial intelligence and machine learning concepts through project-based software development.

PROFESSIONAL EXPERIENCE

Freelance AI & Backend Engineer — Client Projects

2024 – Present

Delivered production-ready AI systems, backend applications, intelligent automation, and business software solutions for local and international clients across e-commerce, finance, and business operations.

- **Shopify AI Marketing Automation** — Engineered 5 production AI automation agents for Shopify store marketing using Zapier and Klaviyo, building customer segmentation logic, personalized lifecycle campaigns, and automated follow-up systems that improved customer engagement.
- **North Steel Financial Modeling** — Built a production financial modeling and forecasting system in Excel, architecting structured revenue, expense, and profit/loss analysis pipelines that delivered decision-ready reporting for management.
- **E-commerce Platform, French Client** — Developed backend architecture for a production e-commerce platform as part of a two-developer team, owning Django REST API development and PostgreSQL database design in an international collaboration.

Python Developer Intern — SoftBeck Limited

June 2025 – August 2025

- Developed backend services using Python and FastAPI, building and maintaining production-quality REST APIs and automation workflows.
- Integrated Gemini and LLaMA APIs to power AI-driven market research features, extending backend functionality with LLM-based analysis.
- Reviewed, tested, and refactored production codebases to strengthen code quality, maintainability, and reliability.

KEY PROJECTS

AI Resume Screening SaaS — Infra

- Developed a FastAPI and PostgreSQL backend for automated resume parsing, candidate management, and AI-powered job matching.
- Implemented JWT authentication and REST APIs to support secure multi-user access.

- Implemented NLP-based skill extraction using spaCy to improve candidate-job matching accuracy.
- Designed scalable SaaS backend architecture supporting resume uploads, filtering, and analytics.

Multi-Agent Research System

- Built a multi-agent research assistant using LangGraph to orchestrate specialized AI agents for research, analysis, and report generation.
- Designed agent orchestration workflows with Server-Sent Events (SSE) streaming for real-time responses.
- Diagnosed and resolved event-loop and API bottlenecks, improving system reliability and responsiveness.

RAG Knowledge Assistant

- Built a Retrieval-Augmented Generation (RAG) knowledge assistant to answer questions grounded in a custom document knowledge base.
- Diagnosed a silent LLM fallback bug that was degrading response quality, and migrated the system to Groq-hosted LLaMA 3.3 70B for more reliable inference.
- Designed the retrieval and generation pipeline to keep responses accurate and traceable to source documents.

AI Customer Support Agent

- Built an AI-powered customer support agent to automatically handle common customer queries using LLM-based conversational logic.
- Designed the system to integrate with existing support workflows, reducing manual response time for routine inquiries.
- Focused on reliable, context-aware responses to improve customer experience while reducing support workload.

AI Lead Qualification Automation

- Built an automated lead qualification workflow in n8n, integrating Groq-hosted LLaMA 3.3 70B to evaluate and score incoming leads against business criteria.
- Connected the workflow to Google Forms, Google Sheets, and Gmail to automatically capture, qualify, and route incoming leads.
- Reduced manual review effort by automating lead assessment and routing, speeding up response to high-value leads.

AI Deepfake Detection Platform

- Built a deepfake video detection platform using a ConvNeXt-Tiny model for frame-level six-class image classification.
- Containerized the system with Docker and deployed it on Hugging Face Spaces for public inference.
- Applied explainable AI techniques to improve prediction transparency and trust in model outputs.

TECHNICAL SKILLS

Programming: Python, SQL

Backend: FastAPI, Django, Flask, REST APIs, JWT

AI Engineering: LangChain, LangGraph, OpenAI, Gemini, Groq, Hugging Face, spaCy

Databases: PostgreSQL, ChromaDB, Supabase, MySQL, SQLite, Firebase

Automation: n8n, Zapier, Klaviyo

Frontend: Next.js, React, Tailwind CSS, Flutter

Cloud & DevOps: Docker, Git, GitHub, Vercel, Google Cloud

CERTIFICATIONS

- IBM AI Engineering Professional Certificate — IBM (Coursera) (In Progress)
- SQL for Data Science — University of California, Davis (Coursera)
- Python — Kaggle
- Data Science Fundamentals — IBM
- Data Analytics & Business Intelligence — DigiSkills.pk